

TIGER ANALYTICS APP ENGINEERING ASSIGNMENT – ASSUMPTIONS DOCUMENT

Business Assumptions

Assumptions	Impact on Design
Re-uploading a CSV for the same store should update existing records, not create duplicates	Upsert logic in <code>`PricingService.uploadCsv()`</code> checks for existing records before insert
All stores use the same CSV format with 5 columns: Store ID, SKU, Product Name, Price, Date	Single CSV parser with fixed header validation; no per-store format customization

Data Assumptions

Assumptions	Impact on Design
Store IDs follow a consistent format (e.g., S001, S002)	<code>`storeId`</code> field is stored as keyword type in ES for exact-match filtering
SKUs are alphanumeric identifiers (e.g., SKU-1001)	<code>`sku`</code> field is stored as keyword type in ES for exact-match filtering
Product names may contain accented characters (e.g., Café, Müsli)	ES <code>`product_name_analyzer`</code> with <code>`asciifolding`</code> filter handles accent-insensitive search
Prices are positive decimal numbers in a single currency per CSV	<code>`BigDecimal`</code> in Java, <code>`DECIMAL`</code> in MySQL, <code>`double`</code> in ES; no multi-currency conversion
Dates are in <code>`yyyy-MM-dd`</code> format	CSV parser strictly validates date format; rejects rows with other formats (e.g., dd/MM/yyyy)
CSV files have a header row as the first line	Parser reads first row as headers and validates against expected column names

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Infrastructure Assumptions

Assumptions	Impact on Design
No authentication/authorization is implemented in the current version	All API endpoints are publicly accessible; Spring Security can be added as a future enhancement
File uploads are small to medium (up to 10,000 rows per CSV)	In-memory CSV parsing is sufficient; no streaming or chunked processing needed

Scope Assumptions

Assumptions	Impact on Design
Single-tenant application — all stores share the same database and ES index	No tenant isolation; `storeId` field differentiates data per store
Edit is limited to Product Name, Price, and Date — Store ID and SKU are immutable	Edit dialog renders Store ID and SKU as readonly fields to preserve the natural key